

4.1 Types of softwares and interrupts (question bank, public)

1. System software vs application software

Question

A user has both system software and application software installed on their computer.

Describe the difference between system software and application software.

Give an example of each software in your answer.

[4]

Two from:

- System software provides services that the **computer** requires
- whereas application software provides services that the **user** requires.

One from system software: **Utility software** (or by example - defragmentation software / antivirus / firewall) or **Operating system**.

One from application software: Any suitable **application** (e.g. word processor / web browser / video-editing software).

Question

A student uses both system software and application software on their computer.

(a) Give **two** examples of system software.

[2]

(b) Give **three** examples of application software.

[3]

(c) Describe the difference between system software and application software.

[2]

This is just a rephrasing of the last question

(a) e.g.

Operating system / Utility software (by example).

(b) e.g.

Word processor / Spreadsheet / Database / Web browser / Image or photo editor / Video editor / Email client / Game / Social media app.

(c)

- System software **manages/maintains** the **hardware** and **software**.
- Application software allows the **user** to perform tasks.

2. Operating systems

Question

State **four** functions of an operating system.

[4]

- Managing files
- Handling interrupts
- Providing an interface
- Managing peripherals and drivers
- Managing memory
- Managing multitasking
- Providing a platform for running applications
- Providing system security
- Managing user accounts

[This is listed in the syllabus.](#)

Question

Handling interrupts is one function of an operating system.

Give **two** functions of an operating system that relate to memory.

[2]

e.g.

- Memory management
- Multitasking

Question

A mobile telephone has an operating system.

Describe the purpose of the operating system.

[3]

- It **performs** the basic functions of a computer
- It **manages** the hardware
- It provides a platform to run software
- It provides a user interface
- It performs tasks such as (any example of function of an operating system)

Question

Each employee in a finance company has a computer.

Each computer has an operating system. One function of the operating system is to manage memory.

Describe the role of the operating system in managing memory.

[3]

e.g.

- To make sure memory is used **efficiently**
- **Allocates/deallocates** memory to processes
- Checks that the processes being actioned have enough memory available
- Moves data to and from memory/between memory and storage
- Makes sure that two processes do not try and access the same memory location
- Creates a **memory partition**
- Creates **virtual memory**
- Transmits **pages** between **RAM and VM**

Question

A computer has an operating system (OS).

Complete the table by writing each missing OS function name and description.

Function name	Description
managing memory	
.....	allows application software to run on the computer
managing peripherals	

[3]

Function name	Description
managing memory	Examples: • allocates memory to processes • prevents two processes accessing the same memory
platform for running applications	allows application software to run on the computer
managing peripherals	Examples: • allocates data to buffers • transmits data to hardware • receives data from hardware

Question

A computer has an operating system.

The operating system provides several functions.

(a) Tick (✓) **one** box to show which of these is **not** a function of the operating system.

- | | | |
|----------|------------------------|--------------------------|
| A | managing files | ☐ |
| B | providing an interface | ☐ |
| C | handling interrupts | ☐ |
| D | loading the bootstrap | ☐ |

[1]

(b) Identify **three** other functions of an operating system and describe the purpose of these functions.

[6]

(a) D

(b) **One** mark for identifying a function. **One** mark for a matching description.

For example:

- Memory management
- ...managing what gets allocated where in memory
- Managing peripherals and drivers
- ... managing the communication between any input and output devices that are connected to the computer
- Multitasking
- ... managing the process of **switching** between tasks that are being carrying out
- Platform for running applications
- ... allowing communication between the applications software and the hardware
- System security
- ... proving features such as username and password
- User accounts
- ... allowing multiple user accounts to be created on a computer

You should ensure you are able to describe each of the above functions.

Question

Software is installed on a computer to manage files, memory and multitasking.

(a) State the name of the software that can do these tasks.

[1]

(b) Give **three** tasks that the software allows the user to do to manage files.

[3]

(c) Describe what is meant by managing memory.

[2]

(a) Operating system

(b)

- Create a file
- Copy a file
- Open a file
- Close a file
- Move a file
- Delete a file
- Rename a file
- Save a file
- Sort files

(c) e.g.

- Keeping track of the status of each memory location
- Managing the movement of data to and from RAM
- Checks that processes have enough memory located to them
- Makes sure that two processes don't try to access the same memory location
- Manage the transfer of pages between virtual memory and RAM
- Allows multitasking

3. Running of applications (role of hardware, firmware, operating system)

Question

A robot is used to move car parts in a factory.

The robot has firmware.

State what is meant by firmware.

[1]

- Permanent **instructions/software** that are programmed into/stored in the ROM.
- **Instructions/software** that allow hardware to be controlled/managed.
- **Instructions/software** that provides the operating system with a platform to run on.

Question

A computer has an operating system.

Give the name of the set of instructions that are provided to the operating system to allow it to run.

[1]

- Firmware
- Bootloader
- BIOS
- Bootstrap

Question

A computer needs firmware and system software to operate.

(a) State the purpose of firmware.

[1]

(b) Give **one** example of firmware.

[1]

(a)

- **Permanently** store instructions (in ROM)
- Stores instructions to **boot up/start up** the computer
- Provides the **operating system** with a platform to run on
- Controls/manages/allows **communication** with hardware
- **Store** instructions **securely** (to stop them being easily corrupted)

(b) e.g.

- Bootstrap
- Bootloader
- BIOS
- Operating system (in embedded system)
- Programs (in embedded systems)

Question

Complete the statements about different types of software.

Use the terms from the list.

Some of the terms in the list will **not** be used. You should only use a term once.

application, assembly language, bootloader, central processing unit (CPU), firmware, hardware, operating, output, system, user

_____ software provides the services that the computer requires; an example is utility software.

_____ software is run on the operating system.

The _____ system is run on the firmware, which is run on the _____.

[4]

system, application, operating, hardware

Question

A computer uses a bootstrap.

Tick (✓) **one** box to show the part of a computer of which the bootstrap is an example.

- | | | |
|----------|----------------------|--------------------------|
| A | application software | ☐ |
| B | firmware | ☐ |
| C | hard disk drive | ☐ |
| D | MAC address | ☐ |

[1]

B

4. Interrupts

Question

One function of the operating system is to manage multitasking.

A signal is sent within the computer to allow multitasking to occur.

State the name of this type of signal.

[1]

Interrupt

Question

A robot is used to move car parts in a factory.

Interrupts are generated when the robot is operating in the factory.

Give **three** examples of interrupts that may be generated when the robot is in operation.

[3]

e.g.

- user input // by example
- encountering an obstacle
- emergency stop
- hardware error
- software error
- battery low
- component overheating
- two processes accessing the same location

Question

A computer has an operating system.

One function of the operating system is handling interrupts.

(a) Describe the role of the operating system in handling interrupts.

[2]

(b) Hardware and software interrupts are two types of interrupt.

(i) Give **two** examples of a hardware interrupt.

[2]

(ii) Give **two** examples of a software interrupt.

[2]

(a)

It assigns a priority to an interrupt.

It uses an interrupt handler / interrupt service routine to process the interrupt.

Maintains/organises the interrupt queue

(b)(i) e.g.

- key press on a keyboard
- button clicks on a mouse
- printer out of ink

(b)(ii) e.g.

- division by zero
- two processes trying to access the same memory location

Question

A printer runs out of ink and needs a new ink cartridge. An interrupt is sent and a message is generated to notify the user that the ink cartridge is empty.

Complete the paragraph about the use of an interrupt in this process.

Use the terms from the list.

Some of the terms in the list will **not** be used. You should only use a term once.

binary, clock, computer, data, error, fetch–decode–execute cycle, hexadecimal, higher, interrupt queue, interrupt service routine (ISR), lower, printer, priority level, secondary storage, sorting, transmission

The sends an interrupt to the The interrupt is given a The processor completes its current and checks the to see if there is an interrupt with priority than the current task. If there is, it stores the current task and fetches the interrupt. It then calls the

..... , which is a sequence of instructions that handles the interrupt. The interrupt is handled, which generates a message to the user to notify them that the ink cartridge is empty.

[7]

- printer
- computer
- priority level
- fetch–decode–execute cycle
- interrupt queue
- higher
- interrupt service routine (ISR)

Question

An interrupt is a type of signal that is used in a computer.

Describe how interrupts are used when a key is pressed on a keyboard.

[5]

- A key press generates an interrupt.
- The interrupt is given a priority...
- ...and sent to the CPU.
- The interrupt is placed in a queue.
- The CPU pauses its current task to check the queue and service the interrupt...
- ...using an interrupt service routine (ISR).
- If the key press has the highest priority, the interrupt is processed.

Question

A computer has an operating system (OS). The OS handles interrupts.

(a) State the purpose of an interrupt.

[1]

(b) Identify **one** input device and **one** example of an interrupt that it can generate.

[1]

(c) A processor is currently executing a program. The processor receives an interrupt. Explain how the processor manages the current program and the interrupt.

[7]

(a) To indicate that something requires the attention of the **processor/OS/CPU**

(b) **One** mark for input device **and** matching interrupt:

e.g.

- Keyboard
- Key pressed
- Mouse
- Mouse moved//button clicked

(c)

- Interrupt is given priority
- ...and placed in interrupt queue
- Processor finishes current **FE cycle** for program
- Processor checks interrupt priority queue // processor checks for higher priority interrupt than program/process
- If lower priority processor runs next FDE cycle for program/process // if lower priority processor continues with program/process
- (if higher priority) processor **stores** current process/registers on stack
- Checks source of interrupt
- ... and calls the appropriate ISR
- ISR handles/resolves interrupt
- If there is another higher priority interrupt (than process) then repeat
- ... (otherwise) processor retrieves content of stack/registers/previous process (to continue with process from program)

The only long process you need to have memorised for this chapter. Be sure you know what component does what (the **processor** checks priority, calls **ISR**, and the **ISR** handles the interrupt, not the processor).

5. Synthesis

These aren't necessarily harder, but may draw from knowledge across Topic 4.1 or beyond.

Question

Write down the correct term for each description below.

1. A collective term for the physical components of the computer system.
2. A type of software that provides services that the user requires and allows the user to perform tasks on the computer.
3. A type of software that manages the main functions of the computer, including managing files and managing memory.
4. A type of software that is stored in the read-only memory (ROM). It includes the basic input/output system (BIOS) and the bootloader.

hardware
application software
operating system
firmware

Question

The table contains **four** descriptions about a computer system.

Complete the table by writing the correct term for each description.

Term	Description
.....	A collective term for the physical components of the computer system.
.....	A type of software that provides services that the user requires and allows the user to perform tasks on the computer.
.....	A type of software that manages the main functions of the computer, including managing files and managing memory.
.....	A type of software that is stored in the read only memory (ROM). It includes the basic input output system (BIOS) and the bootloader.

[4]

Term	Description
hardware	A collective term for the physical components of the computer system.
application software	A type of software that provides services that the user requires and allows the user to perform tasks on the computer.
operating system	A type of software that manages the main functions of the computer, including managing files and managing memory.
firmware	A type of software that is stored in the read only memory (ROM). It includes the basic input output system (BIOS) and the bootloader.

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